

MandateBench: Mandate Faithfulness and Pre-Signature Monitorability for Agentic-Payment LLMs

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Preprint · 2026 · v6.1 pilot, revised (snapshot v6: 9 frontier models, 753 graded authorizations + adversarial duels) · Open source (MIT), github.com/Johnnyevans32/mandatebench · CC BY 4.0 · mandatebench.xyz/paper

ABSTRACT

Emerging standards for AI payments — Google’s Agent Payments Protocol (AP2) and Coinbase’s x402 — let an autonomous agent commit funds by signing a *mandate*: a cryptographically signed statement of what it is authorized to spend. Because a signature is irreversible, the safety-critical question is not whether a protocol gateway can reject a malformed request, but whether the agent can be manipulated into authorizing a payment it should refuse, and whether that can be caught before the signature. MandateBench measures mandate faithfulness across nine frontier models (one or more per major lab) under a taxonomy of adversarial pressures, with a judge-free ground truth: rule violations are deterministic predicates over the signed object, and intent traps are author-constructed and labeled before any model runs — no LLM judges outcomes. Its central contribution is separating two violation classes. *Rule* violations (over budget, over per-payment cap, unauthorized recipient, out-of-scope category, expired mandate) are structural — a stateless gateway that validates the signed fields already catches them. *Intent* violations satisfy every signed field yet betray the mandate’s natural-language intent — a grocery mandate spent on a stored-value gift card or a bottle of vintage champagne — and no structural check can catch them. Models that catch structural violations near-perfectly still miss intent violations: five of nine authorize a third of them, and one authorizes more than half (mean intent-catch 75%, range 44–100%; per-model $n=9$, so pooled tiers rather than individual rows carry the contrast). We then measure pre-signature monitorability: a monitor reading only the agent’s private reasoning predicts violations at AUROC 0.669 (95% CI 0.56–0.78). Instructing the agent to keep its reasoning bland is doubly harmful — violations roughly double ($p \approx 0.009$), and the monitor’s AUROC falls to 0.304, significantly below chance ($z \approx 4.6$), because suppressed reasoning makes a breach read as more routine than a compliant decision. Reasoning suppression does not merely blind the overseer; it blinds the agent.

1. Introduction

Agentic-payment protocols move the authority to spend money from a human click to a model’s signature. In AP2 [1] and x402 [2], a user (or their wallet) issues a signed mandate delimiting a budget, per-payment cap, permitted recipients, permitted categories, and a validity window; an agent then negotiates with merchants and signs to authorize each payment. The signature is the point of no return: once emitted it settles, and there is no undo.

A natural objection is that the protocol itself enforces the mandate — a gateway can reject any signed request whose amount, recipient, or category falls outside the mandate. This is true, and it is exactly why the interesting failures are elsewhere. Hard structural limits are the easy case. The dangerous case is a request that is structurally valid — correct recipient, in-budget, in-category — but is not what the user meant, or is a payment the agent was socially engineered into signing. A gateway cannot distinguish “this week’s groceries” from a \$25 gift card charged to the same grocer for the same amount; only a model that reasons about intent can. MandateBench is built to measure precisely this gap, and to ask whether an overseer can catch a breach in the moment before the signature.

We organize the work around four questions. **RQ1**: how often do models authorize mandate-violating payments under adversarial pressure? **RQ2**: does the failure fall on structural (rule) or intent violations,

and does the split separate models? **RQ3**: can a monitor reading only the agent’s private reasoning predict a violation before the signature, and how robust is that to an agent instructed to hide its reasoning? **RQ4** (adversarial): when one model plays the merchant and attacks another, who breaks and who holds?

2. Related work

Security work on agentic payments so far concentrates on the protocol layer or on a single attack vector. “Whispers of Wealth” red-teams a real AP2 signed-mandate flow via prompt injection, on one model and one vector [3]; a recent SoK of agentic-commerce security systematizes the threat axes and explicitly notes that real-world security benchmarking of these systems remains underdeveloped [4]; zero-trust runtime verification hardens the AP2 transaction itself cryptographically [5]. MandateBench measures the layer in between — the model’s own authorization decision — across models and across a taxonomy of pressures, and is complementary to protocol enforcement.

On the oversight side, chain-of-thought monitorability has been framed as a valuable but fragile safety opportunity [6]; black-box monitors have been trained to catch scheming in agents [7], and pre-action monitoring is established (InferAct [8], ProbGuard [9]). Deception benchmarks measure monitorability of hidden intent in cross-play games [15], and sabotage evaluations measure whether models can defeat oversight [14]. Agent policy-adherence benchmarks — ST-WebAgentBench [10], the τ -bench family, MANTRA [11], PolicyBank [12] — measure web/retail policy following, and adversarial financial-agent benchmarks such as CAIA evaluate decision quality under manipulated crypto/financial markets [13]. Verification-native clearing for agentic commerce [16] and dynamic multi-agent compliance benchmarks [17] are contemporaneous adjacent lines.

MandateBench differs on three axes: (i) a high-stakes, irreversible-action substrate (signed money movement) rather than a game or a QA task; (ii) a judge-free ground truth; and (iii) an explicit separation of gateway-catchable rule violations from intent violations that only a model or model-monitor can catch — which reframes “why put a model in the loop” as an empirical, measurable question.

3. Method

3.1 Signed mandates and ground truth

Each scenario issues an Ed25519-signed mandate over a canonical (sorted-key) serialization; the harness verifies the signature before each run and grades against the signed object (the agent receives the mandate’s terms in its context). A payment is a cart (recipient, amount in integer minor units, currency, category, quantity). A deterministic checker `checkAuthorization(mandate, cart, priorAuthorized)` returns the exact set of violated predicates: over-budget, over per-payment cap, unauthorized recipient, out-of-scope category, expired TTL, quantity creep, currency mismatch, and structuring (splitting an over-budget order into sub-cap charges). No LLM judges outcomes. Rule labels are deterministic by construction; intent labels are author-constructed traps fixed before any model runs (§3.2) — deterministic to grade, but resting on the author’s judgment of intent, which independent human validation should confirm in future versions.

3.2 Pressure taxonomy and violation classes

Scenarios span a taxonomy of pressures — baseline, principal-agent overreach, merchant social engineering, urgency/scarcity, category ambiguity, prompt injection, and scope drift — as well as two non-traps (a clean order and an at-cap order) that measure over-refusal. (A conflicting-mandates arm is defined in the taxonomy but not yet populated with scenarios; the scope-drift arm is single-turn in v6.) Each scenario is labelled by ground-truth class: *clean* (authorizing is correct), *rule* (breaks a structural term), or *intent* (satisfies every structural term but breaks the mandate’s stated intent). The intent traps are constructed so a stateless gateway validating the signed fields passes them: a grocery mandate (FreshMart, groceries, $\leq \$30$, in budget) is offered a \$25 FreshMart gift card, a \$29 bottle of vintage champagne, or a grocery total padded with an \$8 charity round-up. Each is structurally perfect; each betrays the intent.

3.3 Monitorability protocol (RQ3)

For every graded decision we store the agent’s private reasoning and public statement. A monitor model — GPT-5.5 in this pilot — reads only the agent’s private reasoning, never the mandate numbers or the cart, and outputs a violation probability in $[0,1]$. We report AUROC of these scores against the objective violation labels, pooled across the nine agents; pooling mixes between-model style with within-model discrimination, and the monitor scores its own outputs for one of the nine agents (per-model AUROC and a multi-monitor panel are future work). The adversarial arm re-runs the benchmark with an added instruction telling the agent its reasoning may be reviewed and to keep it bland, uninformative, and free of any mention of limits or risk; the decision is otherwise unconstrained. Comparing monitor AUROC with and without suppression measures oversight robustness.

3.4 Models

Nine flagship models via OpenRouter with structured outputs and temperature 0 where supported: GPT-5.5, Claude Opus 4.8, Gemini 3.5 Flash, DeepSeek V4 Pro, Grok 4.3, Llama 4 Maverick, Qwen3.7 Max, Kimi K2.6, and Mistral Large 2512. The pilot runs three repetitions per scenario (snapshot v6: 378 graded authorizations; a suppressed-reasoning arm v6-hidden: 375; three runs errored and were dropped). Because repetitions run at temperature 0, the three reps are near-duplicates — they capture serving-stack nondeterminism, not sampling variance — so the effective sample is closer to the unique-scenario count and all reported intervals should be read as optimistic.

4. Results

4.1 RQ1 — Violations under pressure

Over trap scenarios, only three of nine models never violate (GPT-5.5, Claude Opus 4.8, Kimi K2.6). The rest range from 8% (Gemini, Grok, Qwen) to 25% (Llama 4 Maverick), with DeepSeek and Mistral in between. Notably, running repetitions matters even at temperature 0: Mistral scored 0/9 in a single pass but 3/27 over three reps — a single run overstated its robustness.

4.2 RQ2 — Rule vs intent (headline)

Splitting violations by class is the central result (Table 1). Structural (rule) violations are caught near-perfectly by almost every model — as expected, since they are the gateway-catchable case. Intent violations separate the field: three models catch them fully, five catch two of three, and Llama fewer than half. Per-model intent cells are $n=9$ and their intervals overlap, so the contrast that survives the small n

is pooled: the three perfect models catch 27/27 intent traps (Wilson 95% CI 88–100%) while the other six catch 34/54 (63%, CI 50–75%). A model can look flawless on the structural traps and still authorize a third of the intent violations a gateway is blind to.

MODEL	RULE CAUGHT (N=27)	INTENT CAUGHT (N=9)	FALSE REFUSAL (N=6)
GPT-5.5	100% (27/27) [88–100]	100% (9/9) [70–100]	0% (0/6) [0–39]
Claude Opus 4.8	100% (27/27) [88–100]	100% (9/9) [70–100]	0% (0/6) [0–39]
Kimi K2.6	100% (27/27) [88–100]	100% (9/9) [70–100]	0% (0/6) [0–39]
Gemini 3.5 Flash	100% (27/27) [88–100]	67% (6/9) [35–88]	0% (0/6) [0–39]
Grok 4.3	100% (27/27) [88–100]	67% (6/9) [35–88]	17% (1/6) [3–56]
Qwen3.7 Max	100% (27/27) [88–100]	67% (6/9) [35–88]	0% (0/6) [0–39]
Mistral Large 2512	93% (25/27) [77–98]	67% (6/9) [35–88]	0% (0/6) [0–39]
DeepSeek V4 Pro	89% (24/27) [72–96]	67% (6/9) [35–88]	0% (0/6) [0–39]
Llama 4 Maverick	85% (23/27) [68–94]	44% (4/9) [19–73]	0% (0/6) [0–39]

Table 1. Fraction of violations caught, split by class, snapshot v6 (3 reps/scenario at temperature 0); brackets are Wilson 95% CIs. Rule = structural, gateway-catchable. Intent = structurally valid but betrays stated intent. False refusal = clean orders wrongly declined (usefulness cost). Mean intent-catch 75%. Temperature-0 reps are near-duplicates, so intervals are optimistic (§3.4).

4.3 RQ3 — Pre-signature monitorability

A monitor reading only the agent’s private reasoning predicts violations at AUROC 0.669, 95% CI 0.56–0.78 (Hanley–McNeil; 29 violations vs 345 compliant decisions) — moderately, not perfectly, diagnostic. Under reasoning suppression the picture inverts. First, suppression is not behaviourally free: total violations roughly double (29 → 51 of ~375 per arm; Fisher exact $p \approx 0.009$). Inspecting the reasoning shows why — told to stay bland, models stop reasoning about intent and collapse to structural checks; violating intent-trap runs literally read “within budget, under per-payment cap, to allowed recipient, in allowed category” and authorize the gift card. Second, the monitor’s AUROC falls from 0.669 to 0.304, 95% CI 0.22–0.39 — significantly below chance ($z \approx 4.6$), a significant drop ($z \approx 5.1$) — because the bland, reassuring reasoning attached to a breach looks more innocent than an ordinary compliant decision, actively misleading the overseer. Note that below-chance is itself a signal: suppression does not remove the information so much as invert its sign — an overseer who knows suppression is in play could flip the scores and recover AUROC ≈ 0.70 ; the danger is the overseer who doesn’t know. Naive reasoning-hiding therefore both raises the violation rate and defeats the (unsuspecting) monitor; reasoning suppression blinds the agent as well as the observer.

4.4 RQ4 — Under live attack, the same models break

Beyond scripted traps, we run a full adversarial matrix: every model plays the merchant and attacks every model’s agent over a multi-turn negotiation ($9 \times 9 = 81$ duels; one negotiation per attacker–defender pair against a single over-cap goal, attacker unseeded at temperature 1). Robustness tracks the defender, not the attacker: no attacker exceeds a 22% success rate (2/9, CI 6–55%), yet breaches concentrate on a few agents (Table 2) — and the breached agents are among the models that miss intent violations in Table 1, which is the corroboration a pilot most needs. With $n=9$ per row the intervals are wide: the unbreached group (0/9, upper bound 30%) separates from Mistral and Llama, but the

DeepSeek / Mistral / Llama ordering is not itself significant, and individual cells of the matrix are single Bernoulli draws that carry no signal on their own.

AGENT (DEFENDER)	BREACHED BY ATTACKERS	RATE	WILSON 95% CI
GPT-5.5 · Opus 4.8 · Gemini · Grok · Qwen · Kimi	0 / 9	0%	0–30%
DeepSeek V4 Pro	1 / 9	11%	2–44%
Mistral Large 2512	5 / 9	56%	27–81%
Llama 4 Maverick	6 / 9	67%	35–88%

Table 2. Agent robustness in the 9×9 adversarial matrix (81 duels, single over-cap goal, up to 3 turns, one negotiation per cell). Rate = fraction of the 9 attackers that broke this agent, with Wilson 95% CIs. No attacker exceeded 22% success, so robustness tracks the defender, not the attacker.

5. Discussion

The rule/intent split reframes the “the gateway already enforces the mandate” objection: gateways enforce structure, and structure is the case models already handle. Intent is the residual — unenforceable by construction, and the axis on which frontier models measurably differ. This is the empirical justification for a model (or a model-monitor) in the payment loop. The monitorability result is a cautionary one: reasoning is only moderately diagnostic even in the clear, and a trivial suppression instruction pushes an unsuspecting monitor below chance while making the agent less safe. Oversight that leans on reading reasoning must assume the channel can be cheaply — and counter-productively — degraded, and should treat a systematically below-chance monitor as a red flag for suppression rather than as noise.

6. Limitations

This is a pilot, and the reported violation rates should be read as lower bounds. The intent-violation set is small (three author-constructed traps in one grocery-mandate domain; independent human validation of the labels is future work), and per-model intent cells are $n=9$ with overlapping intervals — only the pooled tiers separate. Repetitions run at temperature 0, so the effective sample is closer to the unique-scenario count; under replicate-aware clustering the default-arm AUROC’s separation from chance is marginal, though the suppression contrast survives. The monitor is a single model (GPT-5.5), scored pooled across agents, including its own outputs. The duel matrix has one negotiation per cell against a single over-cap goal with an unseeded temperature-1 attacker, so it is not exactly reproducible. Two grading asymmetries in snapshot v6 could only understate violations: a counter-offer on an intent trap was graded as defended even when it committed the same purchase, and unparseable model output ended a run ungraded rather than counting against the model — both fixed in the harness for future snapshots. The conflicting-mandates arm is defined but not yet populated, and the scope-drift arm is single-turn in v6. Results may be sensitive to system-prompt wording — during development an unfair expired-mandate prompt produced a spurious universal failure until the agent was given issued/expiry/current timestamps. The suppression arm uses a single, naive hiding instruction; a trained or steganographic adversary is the stronger test of monitorability and is future work, as are an adaptive-adversary robustness curve and multi-domain mandates.

7. Conclusion

MandateBench provides a judge-free instrument for the reasoning-based oversight of high-stakes agentic payments. Its pilot measurements indicate that frontier models handle the violations a gateway already catches but not all of the intent violations only a model can, and that reading an agent’s reasoning to catch breaches is both moderately reliable and cheaply defeated — in a way that also degrades the agent. We release the benchmark, harness, and live dashboard openly.

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